

# Vision Based Surveillance Robot for Military Applications

Janani K<sup>1</sup>, Gobhinath S<sup>2</sup>, Santhosh Kumar K V<sup>3</sup>, Roshni S<sup>4</sup>, Rajesh A<sup>5</sup>

<sup>1,2</sup> Assistant Professor, Department of EEE, Dr.N.G.P Institute of Technology, Coimbatore.

<sup>3,4,5</sup> UG Scholar, Department of EEE, Dr. N.G.P Institute of Technology, Coimbatore.

<sup>1</sup> mailto:janani@gmail.com, <sup>2</sup> s.gobhinath@gmail.com

**Abstract** - Security and monitoring are extremely important in our current society. An efficient monitoring system aids in the analysis of stranded places where humans are unable to access. We intend to construct a surveillance robot that can conduct both active and passive surveillance, with an autonomous patrol mode and a manual mode that alerts the operator when motion is detected. The surveillance system can travel within the surveillance area, allowing more areas of interest to be viewed with the least amount of surveillance equipment possible on a limited budget. Furthermore, acquiring relevant data becomes much easier than surveillance monitoring is adaptable with numerous sensors like humidity/ digital temperature/ sonic/ air quality/ smoke sensors, etc., that may be used for diverse applications. Using the internet as a communication medium, the robotic car can function as both an autonomous and a manually operated vehicle. This multimodal robot was designed to sense the presence of an enemy, capture it on camera, and broadcast it live to an authorised person. Surveillance plays a big role while we're operating on the border, and there's a robot for that. This paper shows how to build a smart surveillance robot for military use utilising a Raspberry Pi for security. A Raspberry Pi field delivers a wireless command, which is received by an authorised user on a web page, and the robot moves as a result. The Raspberry Pi camera is used to stream the video. The Raspberry Pi is programmed in the Python programming language. The video streamed at a rate of up to 15 frames per second, according to the results of the trial.

**Keywords** - Robots, Computer Vision, Raspberry Pi, IoT and GPS

## I. INTRODUCTION

This proposal objective is to utilise modern technologies, which have undergone a revolution, in the sphere of surveillance, especially for military and research initiatives. This concept can be realised by utilising embedded technologies, which aid in the development of a system capable of entering a remote location without the use of human resources. As a result, embedded systems, which are an integrated system of hardware and programme software that can be controlled remotely, are chosen. It discusses a smart surveillance system that uses a Raspberry Pi 3 and a webcam, as well as sensors for specific applications. The Raspberry Pi 3 is popular because it is adaptable and allows direct access to processor pins as GPIOs, making it ideal for developing your concept. The Raspberry Pi 3 is connected to a webcam to produce vision for the Bot. The raspberry pi uses the information obtained from the

cam to control the bot and complete the desired work. Even while upgraded technologies play an important role in a

variety of industries, defence systems have yet to reap the full benefits of these innovations.

The primary goal of this project is to use improvised approaches in defence, medical, and rescue missions. At this time, trespassers and insurgents are frequently infiltrating our country's restricted zones (across borders), and there is no accurate mechanism for identifying individuals. Humans and animals may be distinguished using image processing, and a tracking ID is provided separately to avoid false alarms for allowed access. The Mines are the most dangerous portion of the battlefield. Physically detecting mines opens up a slew of potential dangers. The employment of a remote monitoring bot minimises the danger of detecting and destroying mines. GPR (Ground Penetrating Radar) is one of the most recent micro technologies used to detect landmines, along with the use of Nuclear Magnetic Resonance Spectroscopy (NMR) and other sound sensors. Controls for the arms are employed. As a surveillance application, it can also be used in rescue efforts. Compact bots capable of entering regions where personal contact is impossible are employed to gather information about those locations at aftermath for the natural and human made disasters. Our system assists for pre-monitoring areas, detecting location to an exact precise point using GPS, live image streaming through camera, and roving over any tough terrain. The bot's data is accessed via IoT, which makes the data more secure and authentic.

## II. PROPOSED STRUCTURE FOR HARDWARE IMPLEMENTATION

The goal of this project is to employ modern technologies, which have ushered in a revolution in the sphere of surveillance, especially for military and research initiatives. This concept can be realised by utilising embedded technologies, which aid in the development of a system capable of entering a remote location without the use of human resources. As a result, embedded systems, which are an integrated system of hardware and programme software that can be controlled remotely, are chosen. It discusses a smart surveillance system that uses a Raspberry Pi 3 and a webcam, as well as sensors for specific applications. The Raspberry Pi 3 is popular because it is adaptable and allows direct access to processor pins as GPIOs, making it ideal for developing your concept. The Raspberry Pi 3 is connected to a webcam to produce vision for the Bot. The raspberry pi uses the information obtained from the cam to control the bot and complete the desired work.

Even while upgraded technologies play an important role in a variety of industries, defence systems have yet to reap the full benefits of these innovations. The primary goal of this project is to use improvised approaches in defense, medical, and rescue missions. At this time, trespassers and insurgents are frequently infiltrating our country's restricted zones (across borders), and there is no accurate mechanism for identifying individuals. Humans and animals may be distinguished using image processing, and a tracking ID is provided separately to avoid false alarms for allowed access. The Mines are the most dangerous portion of the battlefield. Physically detecting mines opens up a slew of potential dangers. The employment of a remote monitoring bot minimises the danger of detecting and destroying mines. GPR (Ground Penetrating Radar) is one of the most recent micro technologies used to detect landmines, along with the use of Nuclear Magnetic Resonance Spectroscopy (NMR) and other sound sensors. Controls for the arms are employed. As a surveillance application, it can also be used in rescue efforts. Compact bots capable of entering regions where personal contact is impossible are employed to gather information about those locations in the aftermath through natural and human made disasters. Our system assists in pre-monitoring areas, detecting location to an exact precise point using GPS, live image streaming through camera, and roving over any tough terrain. The bot's data is accessed via IoT, which makes the data more secure and authentic.

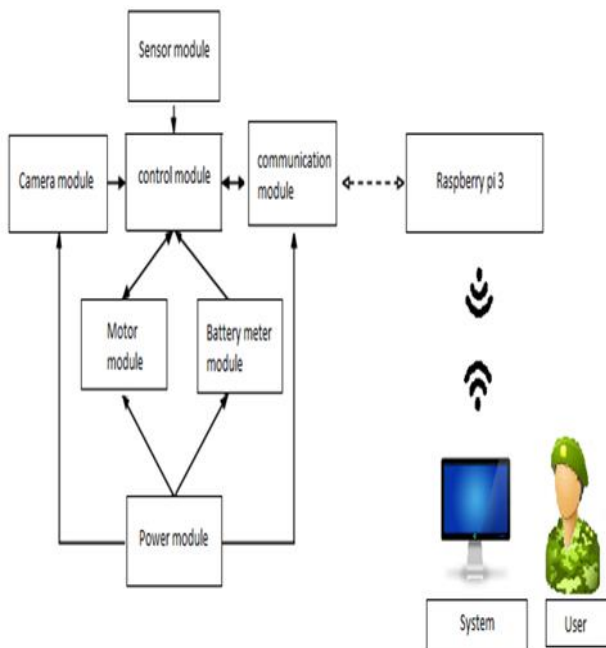


Fig. 1: Block diagram of Proposed Method

#### A. Concept Design:

Through the USB 2.0 ports on the board, the Raspberry Pi 3 Model B with 1 GB RAM is connected to a USB Webcam. The board's 64bit ARM V7 Quadcore processor is capable of processing the 24bit RGB video format captured by the webcam. For our data processing and interpretation, we use a full-size HDMI cable to link the Raspberry Pi 3 to the display. When the system is in a location where HDMI connections are not available, communication can be accomplished via IOT

ideas. The information is delivered to the cloud via the wireless LAN on the board and then retrieved to our display. The Raspberry Pi 3 board is powered by a power bank that can deliver up to 2.4 amps. The DC motors that drive the robot's wheels are driven by 40 pin GPIO input. A 12 V DC supply powers the DC motor. After evaluating the data from the webcam, the Raspberry Pi 3 changed the input to the GPIOs at different times.

#### B. Interfacing raspberry pi 3 with 10mp webcam

Python Idle 2 is the most popular framework for interfacing a Raspberry Pi 3 with a 10MP webcam with a resolution of 640 x 480 pixels. Because an external webcam cannot be directly interfaced in Python idle, Open CV2 must be installed. The Raspberry Pi 3 (Model B) used features an arm 11 core architecture, 1GB RAM memory, and is powered by a 5V, 2A USB connection, with Raspbian OS booting from a microSD card. Frame rates of 40 frames per second and 20 frames per second are available on the webcam interface (VGA). The webcam produces video in 24-bit RGB format. Using a USB connector, the data is send to the Raspberry board.

#### C. Communication of Raspberry Pi with User

For use of Users acknowledgement of any interrupt is to be made by the user there is need of remote accessibility of the board. To implement this concept of Internet of Things is used. Communication using IOT is more secure, fast and can control/monitor the robot from anywhere in the world. The video from the webcam is transmitted to the webserver which is connected to the cloud with specific IP address. This data can be accessed by any PC from the cloud server using the specific IP address. Users can control the bot from the control unit by sending interrupts to the bot through IOT in vice-versa of the method explained above.

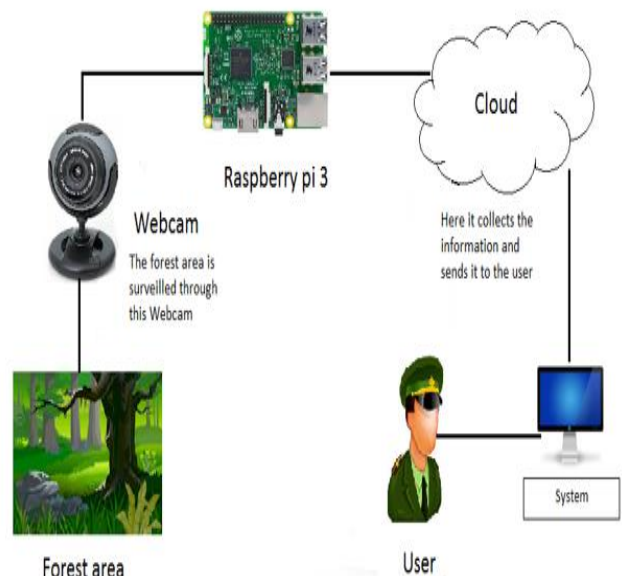


Fig. 2 Diagram for Methodology of communication Protocol

#### D. The Raspberry Pi:

The Raspberry Pi is a low-cost computer the width of a board that connects to a computer monitor or television through a standard mouse and keyboard. [2]. Regardless of its diminutive size, the Raspberry processor is adept of a wide range of tasks and functions similarly to a desktop computer, including Internet browsing, video game play, and word processing. Raspberry Pi, on the other hand it supports an open source languages like as Python and Java, that are compatible with any programming platforms.

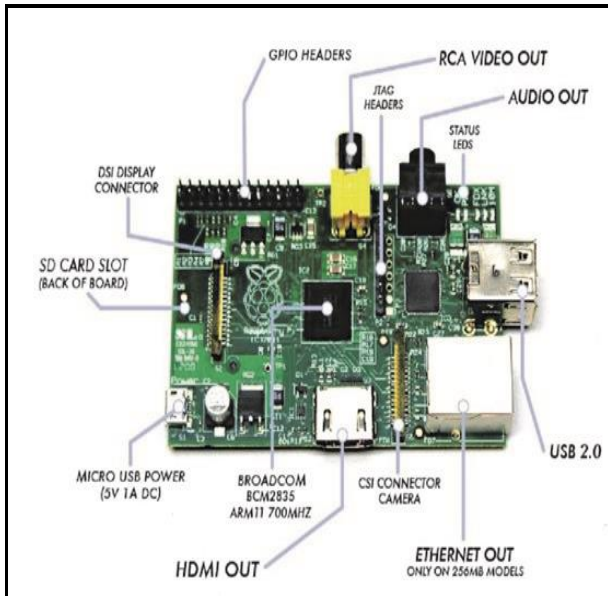


Fig. 3: Raspberry Pi 3 Board Features

Due to its superior performance and specs, Raspberry processor was preferred to other controllers like ARM and arduino for this prototype, allowing high computation applications to be used. The Raspberry Pi 3 is using in this prototype. This prototype was built on a Raspberry Pi 3. The features of the Raspberry Pi 3 are shown in Figure 3. GPIO ports and pins, complete External Memory card slot, audio jack pins and composite video ports, Ethernet pins, camera connector, display connector, High-Definition Multimedia Interface pins, Video Core and 3-Dimensional graphics core [1]. The Raspberry Pi Processor is linked to the Remote Controlled car to provide online video running through the camera output as well as Internet based remote controller of the RC car drive. The Raspberry Pi will be connected to the same network as the monitor, and the output data will be shown in a web portal.

#### E. The Camera:

In this proposed project, the camera is a USB-based web camera [5] linked to the Raspberry Pi. Before installing the web camera, it is tested against a Raspberry processor well-matched USB webcam [3] to guarantee that it will function properly with the Pi. Because the camera is linked to the Pi, the Pi may upload the collected real-time footage to the Internet over Wi-Fi. A web interface can be used to monitor online live video for surveillance or to remotely operate the car measure.

#### F. Raspbian Jessie OS:

The Raspberry Pi runs on the Raspbian operating system. Raspbian is an unofficial Debian Wheezy port with compiler settings tweaked to provide optimised "hard float" code for the Raspberry Pi. For applications that rely heavily on floating point arithmetic operations, this results in much faster performance. Although the main contributors to Raspbian are Peter Green and Mike Thompson, it has greatly benefited from the actively engaging of the Raspberry community who want to get the most out of their gadgets. [7]. Raspbian Jessie is the operating system installed on the Raspberry Pi.

#### G. Python:

Updates to numerous of the Raspbian-bundled applications are among the other unique aspects of Raspbian Jessie. Epiphany web browser, Sonic Pi all has new versions, as well as built-in Python apps. As a result, Python users don't need to install anything to access the file [8]. Python language is a simple and most significant object-oriented programming software that is similar to Java and Scheme, in terms of syntax and readability and makes use of English keywords. To communicate data from GPIO ports to Remote Control Car's motor, this project will use a Python application.

#### H. Overall Activity Flow:

The Raspberry Pi based system's entire activity flow can be broken down into numerous components, as displayed in Figure 4. When the processor is on, it will automatically look for a Wi-Fi connection and interface with to the one that has been assigned to it. The USB based web camera has been instantly start capturing real time camera video. Before being uploaded to a web live server for streaming, the videos will be compressed. Simultaneously, the Pi will await human input or remote commands to controller the RC car's movement.

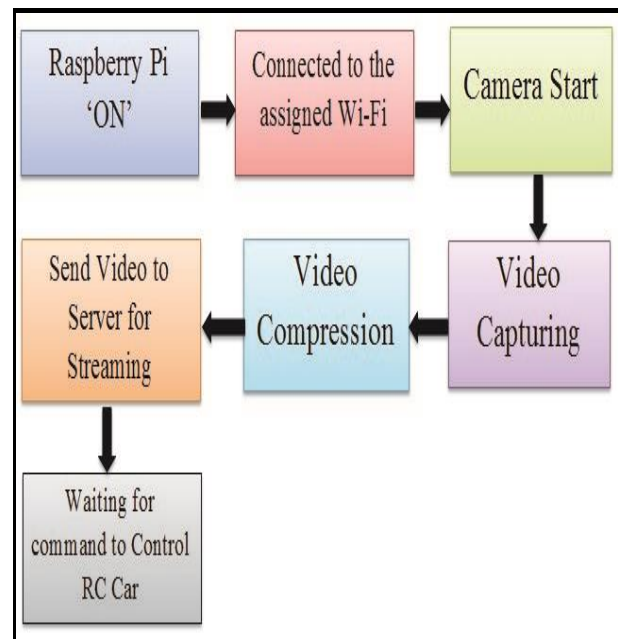


Fig. 4: Overall Activity Flow (Raspberry Pi)

### III. RESULT ANALYSIS & DISCUSSION

The setup and configuration work that went into bringing all of the components together is detailed in this section. Images for the output snapshots are also provided in this section.

#### A. Raspberry Pi Setup

For ensure that every feature functions correctly the first time the Raspberry Pi 3 boots, several devices are required to set it up [2]

#### B. Camera Installation:

Motion, a programme that monitors the visual signal from cameras in order to give live streaming video, will be installed on the USB Webcams. It can detect motion by detecting if a substantial section of the image has moved. Motion may output JPG image format, PNG image format and MP4 video. Later all of the arrangement is finished, the online video sharing will be viewed directly through the browser using a web link [9]. Figure 5 shows a video feed seen in a web browser. The resolution of the image utilised in this project is the default format, is 360pixel width and 240pixel height. MJPG Streamer could be installed and utilised to improve the camera's quality [9]. However, the frame per rate should be properly fix, else movie will stream far too quickly. Another option is to lower the frame rate, alter for video size, and use a high-quality web cam to troubleshoot for video's criteria.



Fig. 5: Interfacing with Raspberry Pi

#### C. OpenCV:

Face detection and recognition can be done with OpenCV, a computer vision framework. The robot uses a camera to perform face recognition, and we manually provide commands to the robot based on that. Unfortunately, the current binary version of OpenCV (version 2.3.x), which can be loaded via apt-get on the Raspbian operating system, is too old to include the facial recognition algorithms needed for this project. However, by obtaining, creating, and installing a newer version of OpenCV, you can access the facial recognition algorithms.

#### D. Webpage Development:

The creation of a web page is a critical component of our effort to control a robot from afar. It will be important to create a single platform from which we will access our robot. We may adjust the direction of the motor and the camera's position as well as monitor the video feed from the web page. It will be important to secure our website against unauthorised access. To keep it safe, we'll keep the IP address a closely guarded secret. When an authorised person wants access to the robot, he will go to that page and input the static IP address of the Raspberry Pi, and the authorised user will be granted access to the robot.

We use 000webhost to create our own website. We created our own website and posted all of the information about our robot on it, as well as our contact information, so that anyone who visits the page may learn about it and leave a remark on it. All of the data will be stored on the server.

#### E. Prototype Development:

The task was done through the Remote Controlled car that delivers online live video sharing for surveillance purposes and can be operated remotely over the web link. Remote Controlled car to be controlled through Raspberry board, which interfaces to web server through Wi-Fi. A Universal Serial Bus camera is attached to the Pi to allow live video broadcasting. A car motor regulator is connected to Raspberry board to control car's movement. A web interface was developed to give the user with both video streaming and an RC controller panel. The user can watch live video while navigating the region by remotely managing the car's movement using this interface.



Fig. 6: Hardware Implementation

A test was performed for local access authorization in which an unauthorized locally connected user could not view the live broadcast and could not control the Remote-Control Car. Unauthorized access is displayed on a web error for such users. In another experiment, the Remote-Control car could only be operated by an authorized user who had the proper id

and password, as well as access to the web browser from anywhere as long as the Pi was turned on.

#### IV. CONCLUSION

In this study, we develop a smart surveillance robot for military use. With the help of this robot, we can determine the current state of the border area in real time without relying on any human source. The surveillance robot provides live streaming video in response to our commands. The Raspberry Pi development board is a low-cost, fully functional computing device that may be used for a wide range of purposes. Camera modules are also reasonably priced and can be used in security systems. We were able to make our product more flexible to future advances by using Python and OpenCV on the Raspberry Pi.. Using this proposed technology, our security officers will have a hand in detecting invaders. As parts of our border areas fall into high altitude locations, this robotic technology can also be deployed in high altitude areas where humans struggle to survive. The proposed robotic system can also be utilised to locate injured people in disasters such as earthquakes, building collapses, and mine areas, as well as serving as a spy robot.

#### REFERENCES

- [1] Shamanth G S and Ashwin Kashyap “face detection using Raspberry pi and python”, National Conference on Product Design, 2016.
- [2] N.V.K. Ramesh, S.C.Gaddam and HemaDhanekula, “Face Recognition Based Attendance Management System with Raspberry PI 2 using Eigen Faces Algorithm”, ARPN Journal of Engineering and Applied Sciences, vol -11, 2016.
- [3] Gobhinath.S and Dineshkumar.V, “Accident and Emergency Informer Using GSM with Voice Support from SPOT Data”, International Journal of Programmable Device Circuits and Systems, Vol 3, no 3, 2011.
- [4] Danwei, “Distributed Compressive Sensing Based Near Infrared and Visible Images Fusion for Face Recognition”, International Journal of Advance Research in Engineering, Science & Technology vol.9, No.4, pp.281- 292, 2016.
- [5] Li Yongqiang and Panjin, “One sample image Recognition algorithm based on improved sub-pattern principle component analysis”, IJSP, IJSPR vol.8, pp.77-84, 2015.
- [6] Jeong-Sik Park, Gil-Jin Jang and Yong-Hoseo, “Efficient Head Pose Determination and its Application to face Recognition on Multi-Pose face DB”, International journal of multimedia and ubiquitous engineering Vol.11.No2 (2016).
- [7] M.Moghaddam and Saeed Meshgini, “Automatic face Recognition via Local Directional Patterns”, Journal of Artificial Intelligence in Electrical Engineering, vol.4, 2015.
- [8] G.Senthilkumar, K.Gopalakrishnan and S.Kumar, “Embedded Image Capturing System Using Raspberry PI System”, International Journal of Emerging Trends & Technology in Computer Science Vol. 3, Issue 2, 2014.
- [9] G.Ghinea, R.Kannan and Suresh Kannaiyan, “Gradient orientation-based PCA Subspace for Novel Face Recognition ”in IEEE Access, Vol 2, 2169-3536 2014.
- [10] M.Angur, Jarman, Samiul Arshad and Mohammed.J.Islam, "An automated Bengali sign language recognition based on finger tip finder Algorithm", International Journal of Electronics & Informatics, 2015.